



Loyola Marymount
University

CMSI 5350/EECE 5150 Machine Learning

Professor Lanyu Shang

Announcement

- Homework 1 due this **Friday**
 - Code -> PostCommit (GitHub)
 - Written part -> Gradescope
 - Don't forget the survey
 - 5-point bonus for early submission by Friday 5pm

Last week

- Mathematical foundations of ML
- Functions and classes in Python
- NumPy
 - NumPy array
 - Index and slice
 - Reshape
 - Concatenate and transpose

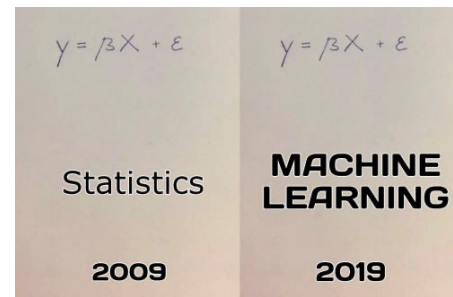


This lecture

- Machine learning
 - Types of learning tasks and examples
 - Mathematical formalization
- Visualization in Python

What is machine learning?

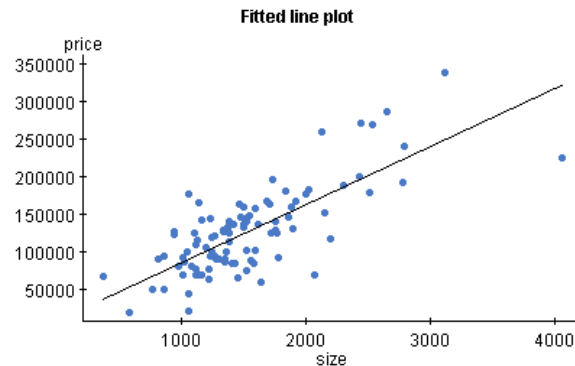
- Wikipedia
 - Machine learning (ML) is a field of study in AI concerned with the development and study of **statistical algorithms** that can learn from **data** and **generalize** to unseen data, and thus **perform tasks** without explicit instructions.
- How does ML differ from statistics?



#10yearchallenge

ML vs. Statistics

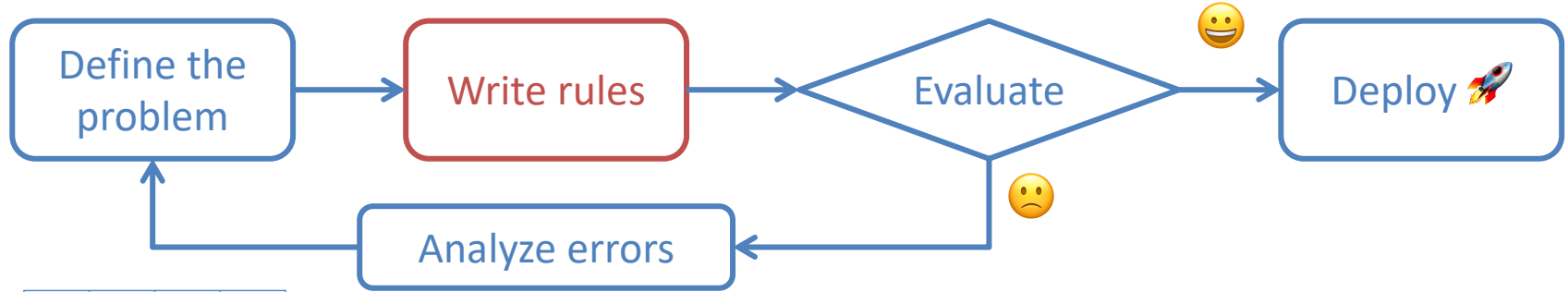
- Example: $y = \alpha x + \beta$
- **Statistics**
 - What is the relationship between x and y ?
 - Coefficient interpretation, hypothesis testing, CI, etc.
- **ML**
 - How well can I predict y from x ?
 - Predictive accuracy, feature engineering, etc.



ML vs. rule-based approach

- Rule-based approach

Pros and cons?



(0,3) 	(1,3)	(2,3) B	(3,3) 
(0,2)	(1,2) B	(2,2) P	(3,2) B
(0,1)	(1,1)	(2,1) B	(3,1)
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Wumpus world

Pros

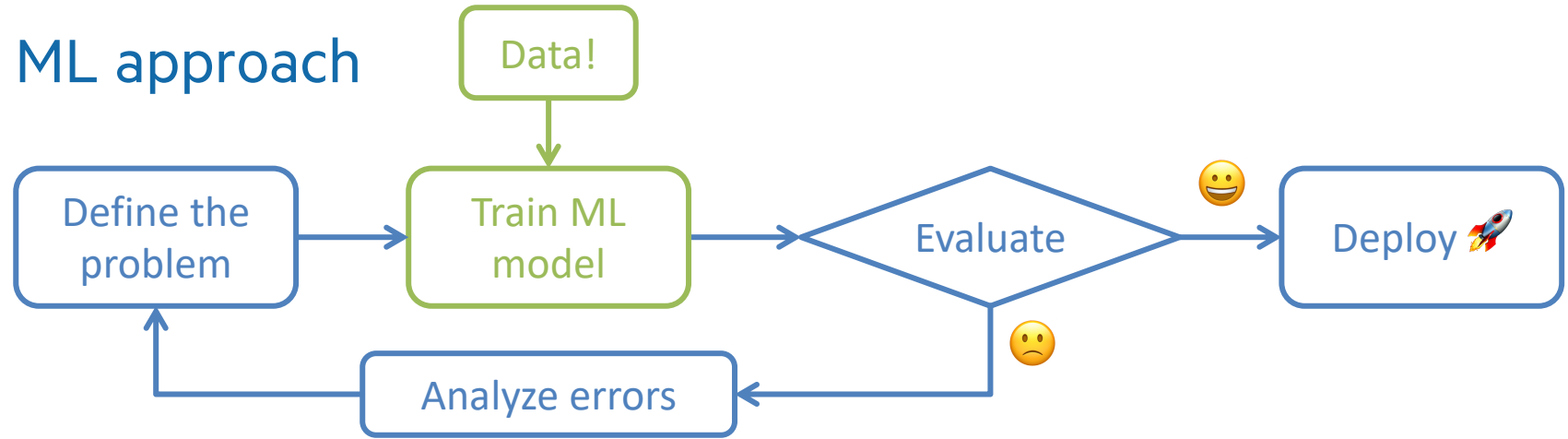
- Domain expertise
- Generally interpretable

Cons

- Labor intensive
- Less generalizable

ML vs. rule-based approach

- ML approach



AlphaZero/MuZero

Pros

- Reduced manual efforts
- Pattern recognition
- Generalizability

Cons

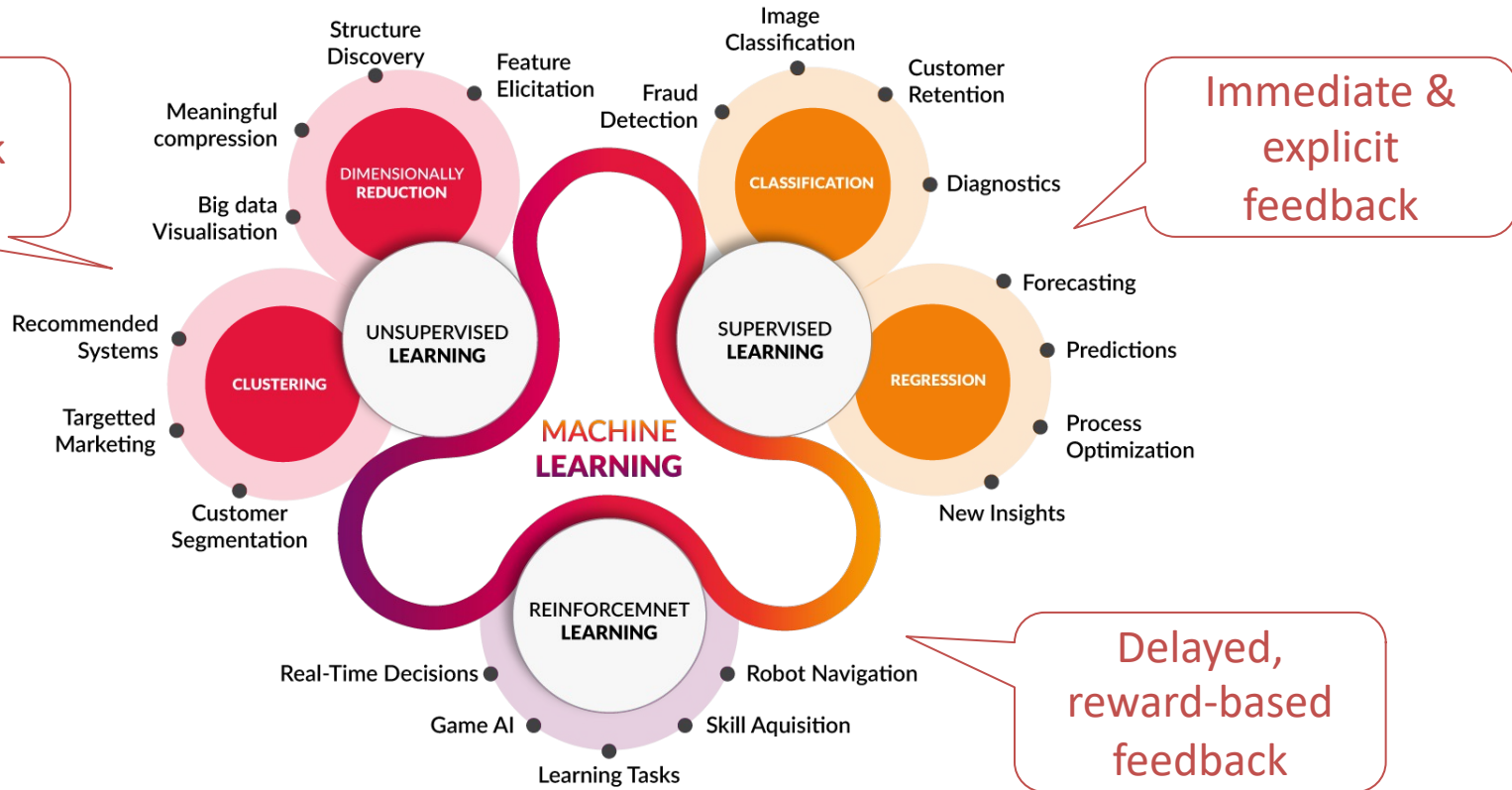
- Data intensive
- Computationally expensive
- Unexpected errors

ML vs. rule-based approach

- ML: if the model predicts well, that's often good enough, even if it's a black box.

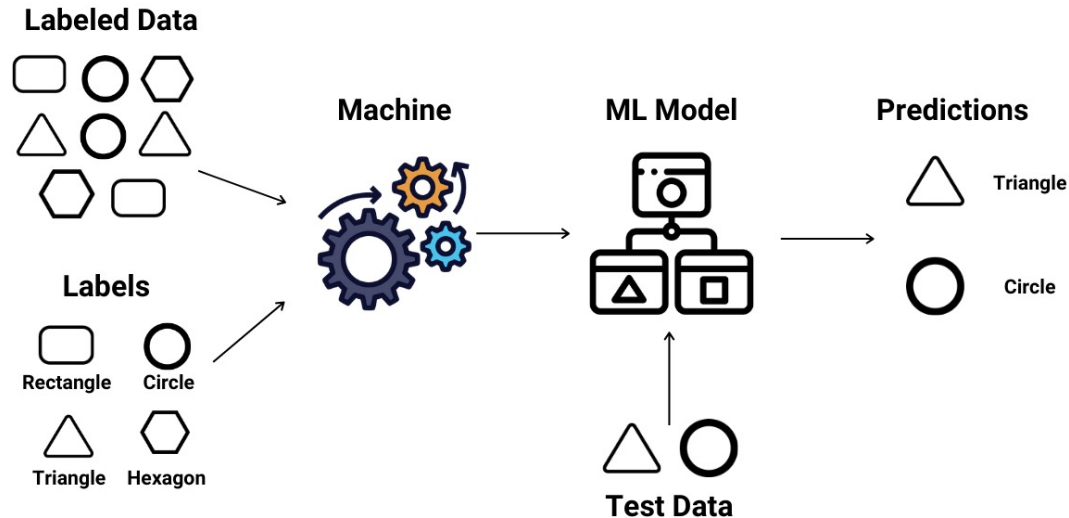


Types of learning – in terms of feedback/label

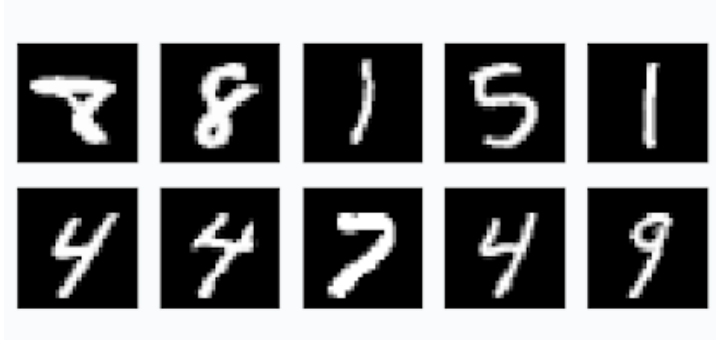


Supervised learning

- Data: Labeled data (input-output pairs)
- Goal: Find a mapping between inputs and outputs

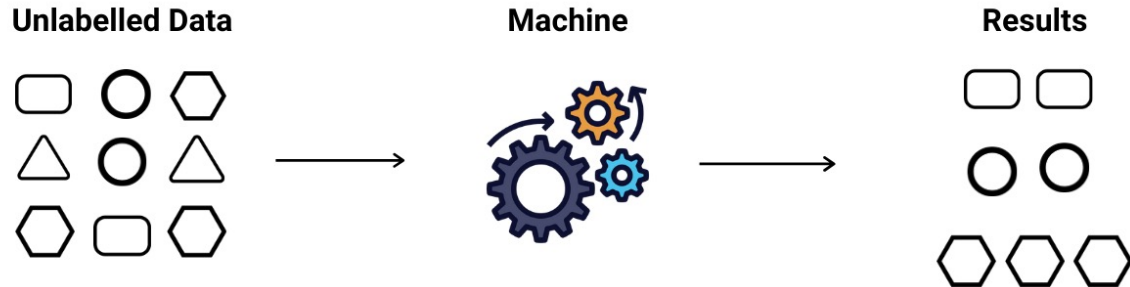


Supervised learning – examples



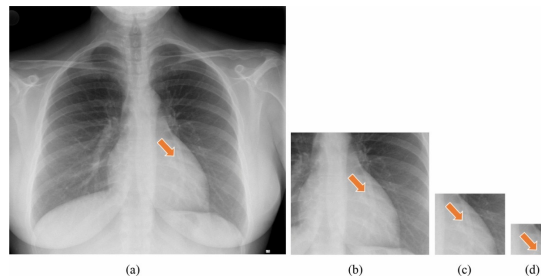
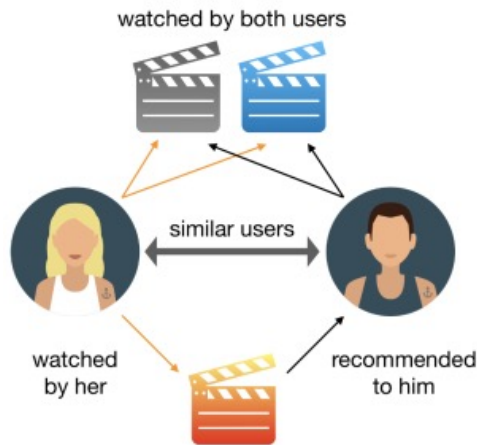
Unsupervised learning

- Data: unlabeled data (no correct answers/targets)
- Goal
 - Find structure/patterns in data
 - Group similar items (clustering), reduce dimensionality

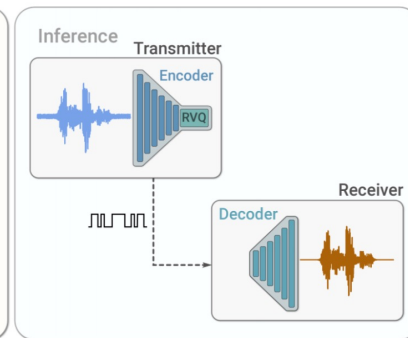
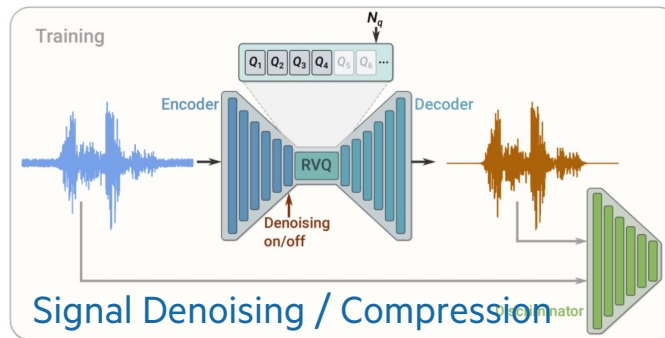


Unsupervised learning – examples

- Examples

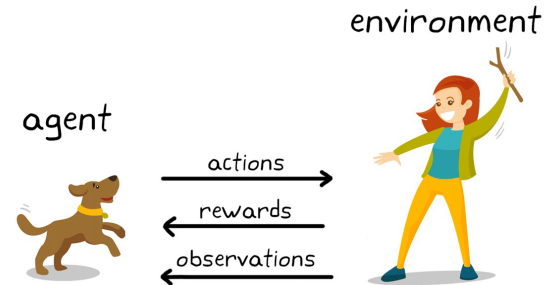
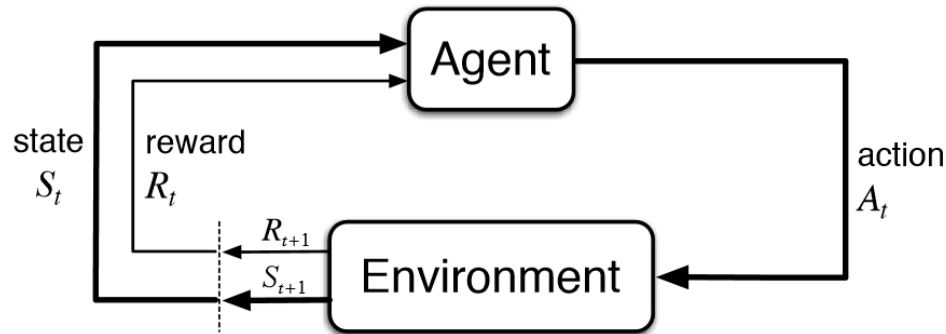


Rare disease diagnosis

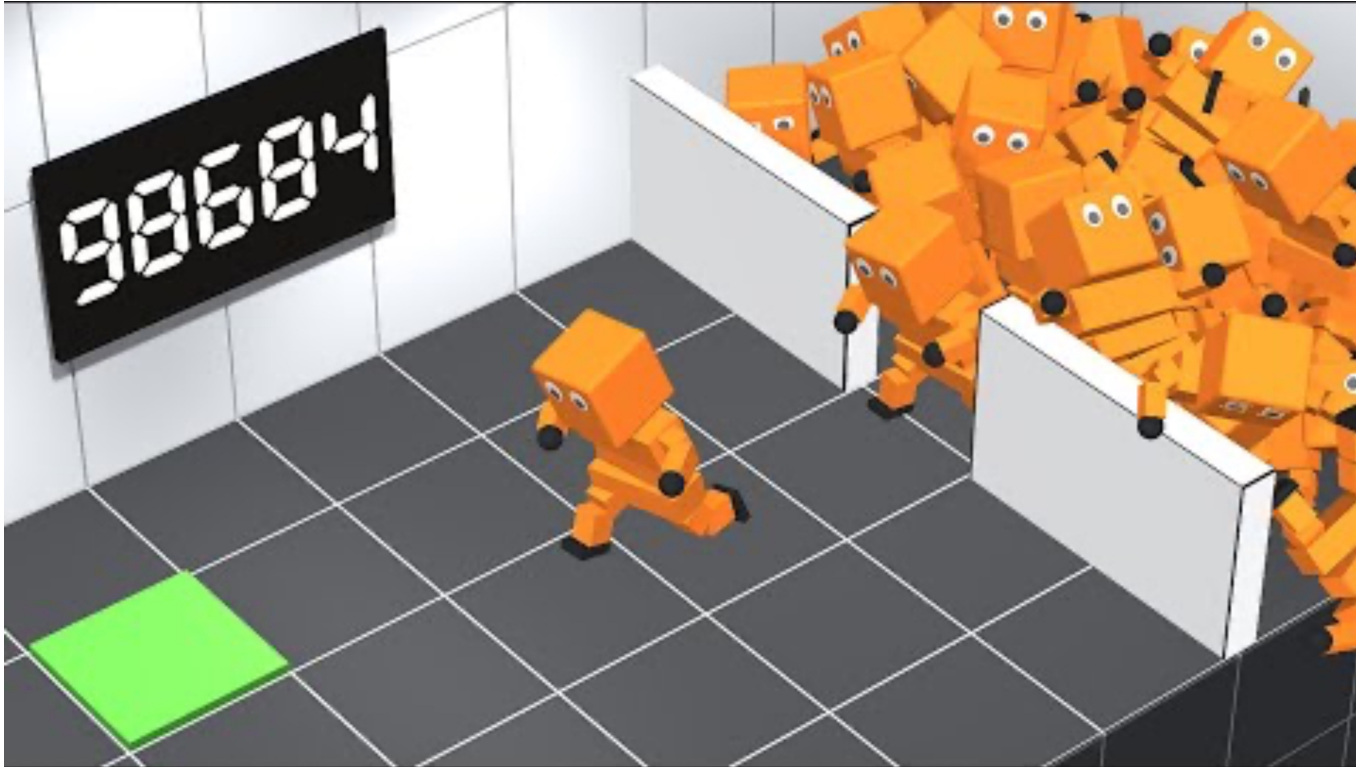


Reinforcement learning

- Data
 - No predefined data
 - Delayed feedback (rewards)
- Goal: maximize rewards by interacting with environment



Reinforcement learning – Albert



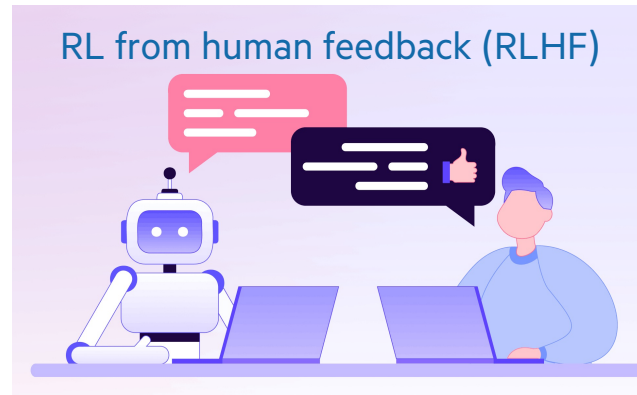
Reinforcement learning – Microduck



Video link: <https://www.youtube.com/watch?v=RAtzEyGBGFU>

Microduck simulator: <https://huggingface.co/spaces/pollen-robotics/microduck-simulator>

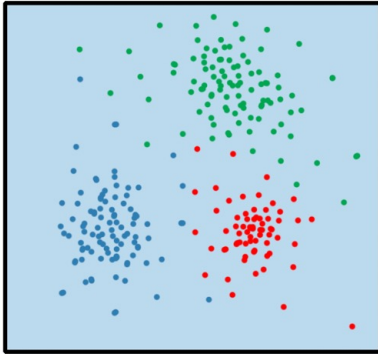
Reinforcement learning - examples



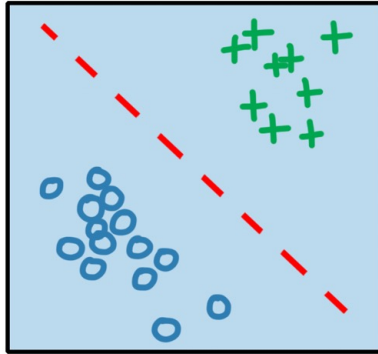
Types of learning

machine learning

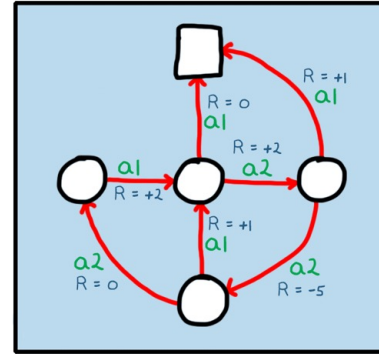
unsupervised
learning



supervised
learning



reinforcement
learning



CMSI 5350/ EECE 5150 (this course)

CMSI 4320/5390

Overview of topics

- Supervised
 - Decision trees
 - KNN
 - Linear regression
 - Logistic regression
 - Perceptron
 - Support vector machine
 - Ensemble methods
- Unsupervised
 - Clustering
 - Dimension reduction
- Deep learning
 - Neural networks/LLM
- Other
 - Bias and fairness
 - Ethics

Colab notebook

VISUALIZATION IN PYTHON